

MACHINE LEARNING WEEK 13

LAB REPORT

Nevin Mathew Thomas

PES2UG23CS381

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Analysis:

1. Dimensionality Justification: Based on the correlation heatmap and explained variance ratio from PCA, why was dimensionality reduction necessary for this dataset? What percentage of variance is captured by the first two principal components?

The correlation matrix shows that many features are weakly correlated or redundant, meaning they do not contribute significant unique information. Applying PCA helps compress these overlapping features into a smaller set of components while retaining the essential patterns in the data. This reduces noise, simplifies the model, and improves computational efficiency without major information loss.

From the explained variance plot:

PC1 $\approx$ 15% PC2 $\approx$ 13–14%

So, the first two principal components together capture about 28–29% of the total variance.

2. Optimal Clusters: Looking at both the elbow curve and silhouette scores, what is the optimal number of clusters for this dataset? Justify your answer using both metrics.

Elbow Method: Sharp decrease up to $k = 3$, then diminishing returns → elbow visible at $k = 3$.

Silhouette Score: 0.39 indicates moderately well-separated clusters, better than scores for higher k values.

Hence $k=3$ is the optimal number of clusters.

3. Cluster Characteristics: Analyze the size distribution of clusters in both K-means and Bisecting K-means. Why do you think some clusters are larger than others? What might this tell us about the customer segments?

The PCA scatter plot shows a dense central region of data and a sparser spread on the edges. K-means naturally assigns more points to dense regions, so clusters with many similar customers become much larger.

The larger clusters (0 and 2) represent the majority customer segments, where customers share similar characteristics (age, balance, call duration, or campaign behaviour).

They form the core audience with consistent patterns.

The smaller group (1) contains customers with distinct or unusual behaviours (e.g., high balance, long/short call durations, or unique contact patterns). This may represent a niche segment.

4. Algorithm Comparison: Compare the silhouette scores between K-means and Recursive Bisecting K-means. Which algorithm performed better for this dataset and why do you think that is?

K-means achieved a silhouette score of 0.39, while Recursive Bisecting K-means achieved a lower score of 0.336. This indicates that standard K-means performed better for this dataset. K-means optimizes all cluster centroids simultaneously, allowing it to find a more globally optimal partition with better separation and tighter cohesion. In contrast, Recursive Bisecting K-means splits clusters one at a time, which can lead to uneven or suboptimal clusters because each split is a local decision rather than a global optimization. As a result, bisected clusters tend to overlap more, reducing the silhouette score.

The clusters show clear customer segments with different financial and behavioural patterns. A large central group represents typical customers suitable for broad marketing campaigns. Smaller, distinct clusters likely capture high-value customers ideal for premium offers and low-engagement or at-risk customers who may need retention or reactivation strategies. Overall, segmentation enables more targeted and effective marketing.

The three coloured regions in the PCA plot represent customer groups with different underlying characteristics such as balance levels, contact frequency, and response behaviour. The boundaries between regions are sharp where customer characteristics differ clearly, causing PCA and clustering to separate them cleanly. In contrast, boundaries become diffuse when customer behaviours overlap, meaning customers share mixed or transitional patterns, which makes the clusters blend gradually instead of forming strict separations.

Feature Correlation Matrix

age job marital education default balance housing loan contact day month duration campaign pdays previous poutcome y

1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75



